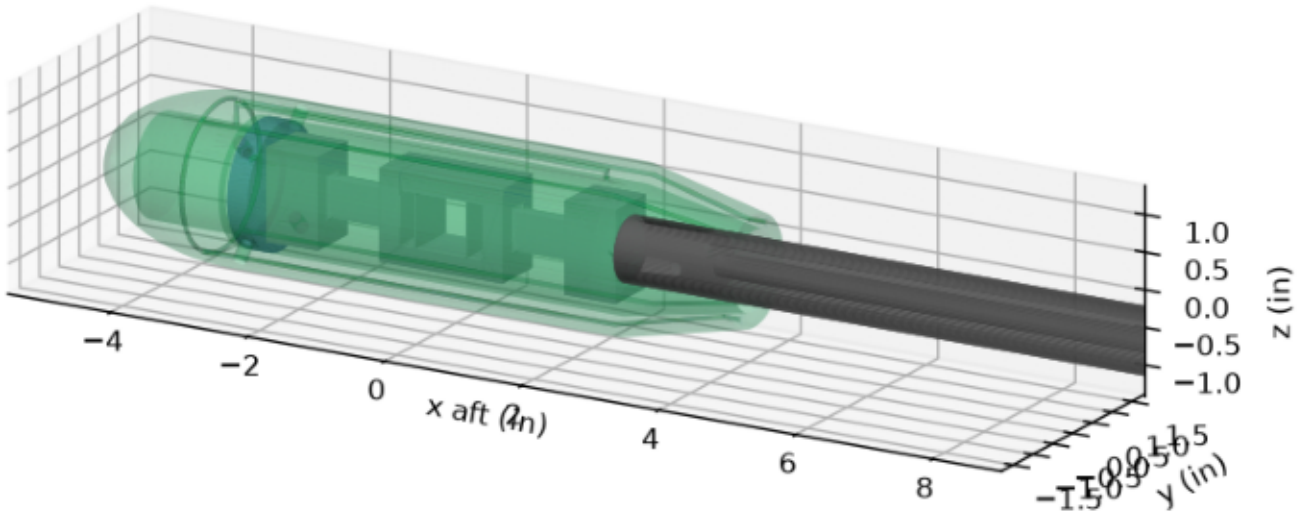


FLIGHT ASSEMBLY (CAD RENDER, NTS)

Flight assembly: instrumented 2.25 in pod (transparent), puck, 6-component balance, sting



NOTES:

1. GAUGE AND WIRE THE BALANCE (24 GAUGES, 6 BRIDGES) BEFORE ASSEMBLY; SEE THE GAUGE LAYOUT DRAWING.
2. PUCK TO BALANCE FRONT FACE: 2 DOWELS + 2 SCREWS. STING ON THE SPIGOT: CROSS PIN + 2 SET SCREWS.
3. WIRES RUN AFT THROUGH THE BALANCE AND THE STING BORE TO J0 AT THE ROLL HOUSING.
4. THE MODEL POD (WT-6C-004) IS 3D PRINTED FROM ITS STEP FILE; THE SHEET GIVES THE FIT AND INSPECTION CHECKS.

UNLESS OTHERWISE SPECIFIED:

DIMENSIONS ARE IN INCHES

TOLERANCES:

.X ±.03 .XX ±.010

.XXX ±.005 .XXXX ±.0005

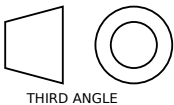
ANGLES ±0°30'

BREAK SHARP EDGES .005-.015

SURFACE FINISH 63 µin Ra MAX

INTERPRET PER ASME Y14.5-2018

UNDIMENSIONED GEOMETRY PER STEP MODEL



PARTS LIST

| ITEM | PART / DWG NO. | DESCRIPTION                        | QTY | MATERIAL / SOURCE  |
|------|----------------|------------------------------------|-----|--------------------|
| 1    | WT-6C-001      | Balance, 6-component (gauged)      | 1   | 7075-T651          |
| 2    | WT-6C-002      | Puck, model interface              | 1   | 6061-T6            |
| 3    | WT-6C-003      | Sting                              | 1   | 4140 28-32 HRC     |
| 4    | WT-6C-004      | Pod, reference model, 2.25 in      | 1   | PC-ABS, 3D printed |
| 5    | -              | Strain gauge 350 ohm, CEA-13 class | 24  | Vishay / bought    |
| 6    | -              | Dowel pin 1/8 x 1/2                | 2   | bought             |
| 7    | -              | SHCS #10-32 x 5/8 (puck)           | 2   | bought             |
| 8    | -              | Cross pin 3/16 dowel (sting)       | 1   | bought             |
| 9    | -              | Set screw 1/4-28 cup point         | 2   | bought             |
| 10   | -              | SHCS #8-32 x 3/8 (pod)             | 4   | bought             |
| 11   | -              | Wire 32 AWG PTFE + J0 DC-37        | 1   | see Electronics    |

NEXT ASSY: WT-6C-400 PITCH SECTOR ASSY

|                                                   |  |     |      |                                  |  |          |  |         |
|---------------------------------------------------|--|-----|------|----------------------------------|--|----------|--|---------|
| PROJECT                                           |  |     |      | DRAWN / DATE                     |  |          |  |         |
| Wind tunnel - motorized 6-component sting balance |  |     |      | Claude (AI-generated) 2026-10-02 |  |          |  |         |
| TITLE                                             |  |     |      | CHECKED                          |  | APPROVED |  |         |
|                                                   |  |     |      |                                  |  |          |  |         |
| ASSEMBLY, FLIGHT (BALANCE + STING + MODEL)        |  |     |      |                                  |  |          |  |         |
| MATERIAL                                          |  |     |      | FINISH                           |  |          |  |         |
| SEE PARTS LIST                                    |  |     |      | SEE PART DRAWINGS                |  |          |  |         |
| DWG NO.                                           |  | REV | SIZE | SCALE                            |  | WEIGHT   |  | SHEET   |
| WT-6C-100                                         |  | A   | B    | NTS                              |  | -        |  | 1 OF 36 |